Cancer Driver Gene Prediction via Explainable Multilayer Graph Neural Networks with Multi-Network Integration and Pathway Enrichment Analysis

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ABSTRACT

Identifying cancer driver genes from the vast landscape of somatic mutations is a central challenge in computational oncology. We build upon and extend the Explainable Multilayer Graph Neural Network (EMGNN), which fuses multi-omics node features with protein–protein interaction (PPI) network topology via a two-stage graph neural network architecture. Our work pursues four objectives. First, we reproduce the benchmark, achieving a mean AUPR = 0.743 (best = 0.753) on the CPDB network (GCN backbone, 18 runs), confirming the published trend. Second, we perform systematic ablation of four architectural improvements—residual skip connections, Batch Normalisation, label smoothing (ε = 0.05), and cosine learning-rate annealing—and find that BatchNorm1d is detrimental in full-batch graph settings (-4.2% AUPR); residual connections and label smoothing are retained as beneficial components. Bayesian hyperparameter optimisation (Optuna, 50 trials) identifies a superior configuration achieving AUPR = **0.8023** (+5.4% over baseline). Third, we extend the framework to all six PPI databases using learnable per-network importance weights, achieving AUPR = **0.8067** and AUROC = **0.9170**, a gain of **+5.9 percentage points** over the single-network baseline. Learned network weights confirm CPDB and Multinet as the most informative databases (weights = 0.210 and 0.201). Fourth, we apply Integrated Gradients attribution—revealing that DNA methylation and gene expression features dominate the cancer signal—and Gene Set Enrichment Analysis via Enrichr, recovering 31 significant Hallmark pathways (FDR < 0.05); the highest-ranked terms include Epithelial Mesenchymal Transition (FDR = 1.6×10^{-17}), PI3K/AKT/mTOR Signalling (FDR = 5.8×10^{-15}), and Apoptosis (FDR = 6.4×10^{-11}). The top predicted genes—TP53, MUC16, CTNNB1, PIK3CA—are all well-established oncogenes or tumour suppressors, validating the model's biological fidelity. These findings demonstrate that multi-network integration, systematic hyperparameter optimisation, and interpretability analysis are complementary strategies for building robust, trustworthy cancer gene predictors.

Keywords: cancer driver gene prediction, graph neural network, protein–protein interaction network, multi-omics integration, gene set enrichment analysis, model interpretability, hyperparameter optimisation

1 Introduction

Cancer arises through the accumulation of somatic mutations in a subset of genes—termed *cancer driver genes*—that confer selective growth advantages to cells. Distinguishing these drivers from the large background of passenger mutations is essential for understanding tumourigenesis, stratifying patients, and selecting targeted therapies [2]. Genome-wide sequencing projects such as TCGA have catalogued thousands of mutations across dozens of cancer types, yet the challenge of prioritising functionally relevant alterations at scale remains largely unsolved [1].

Graph Neural Networks (GNNs) have emerged as a natural framework for this problem because PPI networks provide a biologically meaningful relational structure: genes with similar functions tend to cluster in network neighbourhoods, and disease genes are enriched in specific modules [3]. Early GNN-based approaches such as EMOGI [2] encode each gene as a node with multi-omics features and propagate information through a single PPI network, achieving state-of-the-art precision–recall performance. The more recent EMGNN framework [1] generalises this to a multilayer setting by constructing a *meta-graph* whose nodes aggregate evidence from multiple per-network GNN encoders, and further provides Integrated Gradients explanations at the edge and feature level.

Despite these advances, several open questions remain. First, it is unclear which architectural improvements—residual connections, normalisation, regularisation via label smoothing, adaptive learning-rate schedules—translate to gains on biologically sparse, full-batch graph tasks, where the entire network is processed in a single forward pass. Second, while EMGNN nominally supports multiple PPI networks, the original evaluation is primarily single-network; the extent of the multi-network gain and the role of per-network importance weights have not been systematically quantified. Third, pathway-level interpretability—connecting gene predictions to known cancer hallmarks—has not been fully explored within the EMGNN framework [11].

This work closes these gaps with a systematic study of the EMGNN framework that moves from reproduction to improvement to interpretation. We first establish a faithful reproduction of the original benchmark and then ask: which architectural choices actually help? Our ablation and Bayesian hyperparameter search reveal that BatchNorm1d is *harmful* in full-batch graph settings (-4.2% AUPR), while removing it together with tuned hyperparameters yields AUPR = 0.8023 (+5.4% over baseline). We then show that training across all six available PPI databases with learnable per-network importance weights further improves performance to AUPR = 0.8067 (+5.9%), while the learned weights quantify each database's relative contribution. Finally, Integrated Gradients and GSEA analysis reveal that the model relies primarily on epigenetic (methylation) and transcriptomic (gene expression) signals rather than mutation frequency, and that its top-ranked genes are significantly enriched in canonical cancer hallmarks including EMT, PI3K/AKT/mTOR, and G2-M checkpoint pathways.

2 Research Methodology

2.1 Datasets

We use the six PPI network datasets from the EMOGI benchmark [2]: CPDB (13,627 nodes), IRefIndex (17,013), IRefIndex-2015 (12,129), Multinet (14,398), PCNet (19,781), and STRINGdb (13,179). Each network is stored as a dense adjacency matrix in HDF5 format. To avoid materialising multi-gigabyte dense matrices (PCNet requires $\approx$3 GB as float64), we implement a chunked sparse reading strategy: the adjacency matrix is read in blocks of 2,000 rows, nonzero indices are extracted, and the result is stored as a compressed `.npz` cache alongside the original file. Cached loading reduces total data loading time from ≈96 s to ≈1.3 s for all six networks.

Node features are 64-dimensional multi-omics vectors comprising mutation frequency (MF), DNA methylation (METH), gene expression (GE), and copy-number alteration (CNA), each across 16 TCGA cancer types, with a fixed canonical ordering aligned across networks. Cancer/non-cancer labels and train/val/test masks are loaded from the same HDF5 containers.

2.2 EMGNN Baseline Architecture (Methodology 1)

The EMGNN architecture [1] operates in two stages, described in Section 3. A shared GNN encoder processes each PPI graph independently, and the resulting node embeddings are aggregated through a meta-graph whose nodes represent unique genes across all networks. A second GNN on the meta-graph produces final embeddings that are classified by an MLP. We reproduce the benchmark using the original code with GCN [3], GAT [4], and GIN [5] backbones, training with Adam [12] (lr = 5×10^{-3}, weight decay = 5×10^{-4}), 2000 epochs, and early stopping with patience 250.

2.3 Architectural Improvements (Methodology 2)

We introduce four architectural modifications to the EMGNN baseline and evaluate them via a controlled ablation study on CPDB with a GCN backbone.

2.3.1 Residual Skip Connections

Following He et al. [14], we add element-wise residual connections between consecutive GNN layers for all layers after the first. This improves gradient flow in deeper configurations and stabilises training.

2.3.2 Normalisation Strategies

We optionally apply layer-wise normalisation after each GNN layer, controlled by the `norm_type` parameter with four options.

BatchNorm1d [13] normalises activations over the batch dimension. Through ablation we find that, in the full-batch graph setting where the entire graph is a single "batch", the running statistics are computed over all N nodes simultaneously, providing no meaningful distributional shift correction across training steps. `BatchNorm1d` is therefore set as a configurable but *cautioned* option, disabled in our best-performing configuration.

GraphNorm [16] addresses this limitation by normalising activations within each individual graph instance rather than across a batch. This preserves meaningful statistics in the full-batch setting and is implemented as the recommended alternative to BatchNorm for graph-level tasks.

LayerNorm normalises over the feature dimension for each node independently, requiring no batch or graph structure information. This is the most general option and operates correctly regardless of training batch size or graph partitioning.

Both GraphNorm and LayerNorm are implemented and exposed via `-norm_type graph` and `-norm_type layer` respectively.

2.3.3 Label Smoothing

Hard binary targets are replaced with soft labels:

$$\tilde{y} = y\,(1 - \varepsilon) + \varepsilon / K, \tag{1}$$

where $\varepsilon = 0.05$ and $K = 2$. This reduces over-confidence on the imbalanced cancer/non-cancer dataset and improves calibration.

2.3.4 Learning-Rate Annealing and Gradient Clipping

We use cosine annealing [15] to decay the learning rate from η_0 to $\eta_{\min} = 10^{-5}$, and clip gradients to $\|\nabla\|_2 \leq 1.0$ before each optimiser step. These measures reduce late-training oscillation and prevent exploding gradients in deep configurations.

2.3.5 Feature Engineering Pipeline

A preprocessing pipeline is applied to node features before training: (i) variance-threshold feature selection (threshold = 0.01) removes near-constant features; (ii) Z-score standardisation centres each feature and normalises its variance. The pipeline is fitted on the first network and applied consistently to all subsequent networks to prevent data leakage.

2.3.6 Hyperparameter Optimisation

We use Bayesian optimisation via Optuna (TPE sampler, median pruner) to search over learning rate, hidden dimension, number of layers, dropout, weight decay, normalisation type, label smoothing, and LR scheduler. Fifty trials are run with 2000 training epochs per trial and early stopping (patience = 250). The best configuration is reported in Section 4.2.

2.4 Multi-Network Extension (Methodology 3)

We extend the improved model to simultaneously train on multiple PPI networks and introduce a learnable per-network importance weighting mechanism. A vector $\boldsymbol{\theta} \in \mathbb{R}^K$ is learned jointly with the rest of the model; at each forward pass, the weights $\mathbf{w} = \text{softmax}(\boldsymbol{\theta})$ scale the embeddings of all nodes belonging to each network k, allowing the model to up-weight high-quality networks automatically. We evaluate all six network combinations reported in Table 4, ranging from two to six PPI databases, and train for 2,000 epochs with patience 250 on an RTX 5090 GPU.

2.5 Interpretability Analysis (Methodology 4)

2.5.1 Integrated Gradients Attribution

We apply Integrated Gradients [7] to attribute each prediction to individual node features. Attributions are aggregated across all cancer-label meta-nodes by taking the mean absolute value per feature, producing a global feature importance ranking across the four omics modalities. The `AttributionAnalyzer` class provides a clean API over Captum [?], with separate methods for node-level and gene-set-level attribution.

2.5.2 Gene Set Enrichment Analysis

We apply two complementary GSEA methods over the MSigDB Hallmark 2020 gene set library [10]. First, Enrichr ORA [9] is run on the top-200 predicted genes (by cancer probability), producing FDR-adjusted p-values. Second, pre-ranked GSEA [8] uses the continuous prediction scores to rank all genes and computes normalised enrichment scores (NES). Statistical significance is set at FDR < 0.05.

2.6 Experimental Setup

All experiments are run on an NVIDIA RTX 5090 GPU (CUDA 12.8, sm_120) with PyTorch 2.7 and PyTorch Geometric 2.7. Random seed is fixed to 72 for reproducibility. The Adam optimiser [12] is used throughout. Evaluation metrics are the Area Under the Precision–Recall Curve (AUPR) and the Area Under the ROC Curve (AUROC), computed on the held-out test set of the last specified network (CPDB in all multi-network experiments). For single-network experiments we report both the best result across multiple independent runs and the mean where applicable.

3 Theory and Model Architecture

3.1 Per-Network Graph Neural Network Encoder

Let $\mathcal{G}^{(k)} = (\mathcal{V}^{(k)}, \mathcal{E}^{(k)})$ denote the k-th PPI network with node feature matrix $\mathbf{X}^{(k)} \in \mathbb{R}^{N_k \times D}$, where N_k is the number of genes in network k and $D = 64$ is the multi-omics feature dimension. A shared L-layer GNN encoder produces node embeddings:

$$\mathbf{H}^{(k,l)} = \sigma\Big(\mathrm{GNN}_l\Big(\mathbf{H}^{(k,l-1)},\, \mathbf{A}^{(k)}\Big)\Big), \quad l = 1, \ldots, L, \tag{2}$$

where $\mathbf{H}^{(k,0)} = \mathbf{X}^{(k)}$, $\mathbf{A}^{(k)}$ is the symmetrically normalised adjacency matrix with added self-loops, and σ is LeakyReLU. We support four GNN variants:

GCN [3]: $\mathrm{GNN}(\mathbf{H}, \mathbf{A}) = \hat{\mathbf{A}}\mathbf{H}\mathbf{W}$, where $\hat{\mathbf{A}} = \tilde{\mathbf{D}}^{-1/2}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-1/2}$ is the symmetric normalised adjacency.

GAT [4]: Attention coefficients α_{ij} are learned via a shared attentional mechanism over node pairs, allowing the model to weight neighbour contributions adaptively.

GIN [5]: $\mathrm{GNN}(\mathbf{H}, \mathbf{A}) = \mathrm{MLP}((1 + \epsilon)\mathbf{H} + \mathbf{A}\mathbf{H})$, maximally expressive within the Weisfeiler–Lehman hierarchy.

GraphSAGE [6]: $\mathrm{GNN}(\mathbf{H}, \mathbf{A}) = \mathbf{W}\,[\mathbf{h}_u \,\|\, \mathrm{MEAN}_{v \in \mathcal{N}(u)} \mathbf{h}_v]$, concatenating a node's own embedding with its mean neighbourhood embedding.

When residual connections are enabled, layers $l \geq 2$ add a skip connection:

$$\mathbf{H}^{(k,l)} = \sigma\Big(\mathrm{GNN}_l\Big(\mathbf{H}^{(k,l-1)}, \mathbf{A}^{(k)}\Big)\Big) + \mathbf{H}^{(k,l-1)}. \tag{3}$$

3.2 Meta-Graph Construction

After per-network encoding, a meta-graph $\mathcal{G}^{\text{meta}}$ is constructed. Each unique gene across all K networks receives one meta-node; directed edges connect each per-network copy of gene i (in network k) to its meta-node. Meta-node features $\mathbf{X}^{\text{meta}} \in \mathbb{R}^{M \times D}$ are initialised by assigning each meta-node the multi-omics features of the corresponding gene (from whichever network first provides them), where M is the number of unique genes. This design aggregates cross-network evidence without redundancy and scales linearly with the number of unique genes.

3.3 Learnable Per-Network Importance Weights

In the standard EMGNN, all network encoders contribute equally to the meta-graph. We introduce a trainable weight vector $\boldsymbol{\theta} \in \mathbb{R}^K$, producing normalised network importances:

$$\mathbf{w} = \text{softmax}(\boldsymbol{\theta}), \quad w_k = \frac{e^{\theta_k}}{\sum_{j=1}^{K} e^{\theta_j}}. \tag{4}$$

The embedding of each input node in network k is scaled by w_k before being passed to the meta-graph GNN, allowing high-quality networks to be up-weighted end-to-end during training.

3.4 Meta-Graph GNN and Classifier

A second L-layer GNN (same backbone) operates on $\mathcal{G}^{\text{meta}}$, aggregating cross-network signals over meta-node edges:

$$\mathbf{Z} = \text{GNN}^{\text{meta}}\big(\mathbf{X}^{\text{meta}}, \mathbf{A}^{\text{meta}}\big) \in \mathbb{R}^{M \times H}. \tag{5}$$

Each meta-node embedding $\mathbf{z}_i$ is passed through a two-layer MLP with softmax output to produce cancer-class probabilities $\hat{p}_i$. The model is trained with cross-entropy loss, optionally with label smoothing as in Eq. 1.

3.5 Normalisation in Full-Batch Graph Learning

Standard Batch Normalisation [13] normalises over the batch dimension:

$$\hat{h}_i = \frac{h_i - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}}, \quad \mu_{\mathcal{B}} = \frac{1}{N}\sum_{i=1}^{N} h_i, \tag{6}$$

where $\mathcal{B}$ is the set of all N nodes. In the full-batch graph setting there is no variation across batches, so the running statistics computed during training equal those at inference; the normalisation is effectively a fixed affine transform and provides no regularisation benefit.

GraphNorm [16] instead computes per-graph statistics, subtracting a learnable mean shift $\alpha \cdot \mu_{\mathcal{G}}$ before normalisation:

$$\hat{h}_i = \frac{h_i - \alpha\,\mu_{\mathcal{G}}}{\sqrt{\sigma_{\mathcal{G}}^2 + \epsilon}}, \quad \mu_{\mathcal{G}} = \frac{1}{|\mathcal{V}|}\sum_{i \in \mathcal{V}} h_i, \tag{7}$$

where $\alpha \in [0, 1]$ is a trainable scalar that controls the degree of mean removal. This preserves meaningful within-graph distributional information that `BatchNorm1d` discards in the full-batch regime.

Our improved model supports `norm_type` $\in$ {`batch`, `graph`, `layer`, `none`}, implemented via `BatchNorm1d`, `GraphNorm` (from PyTorch Geometric), `LayerNorm`, and identity respectively. Ablation results in Section 4.2 demonstrate the cost of `batch` normalisation; direct comparison of the remaining options is future work.

3.6 Integrated Gradients for Interpretability

Integrated Gradients [7] satisfies the *completeness* axiom: the sum of all feature attributions equals the difference between the model output at the input and at the baseline. For a gene with feature vector $\mathbf{x}$ and zero baseline $\mathbf{x}' = \mathbf{0}$:

$$\mathrm{IG}_d(\mathbf{x}) = x_d \int_0^1 \frac{\partial F(\alpha \mathbf{x})}{\partial x_d} \, d\alpha, \tag{8}$$

where F is the cancer-class logit and the integral is approximated with 50 Riemann steps. This property ensures that attributions account for the full prediction score, making Integrated Gradients preferable to vanilla gradient methods for model-level interpretability.

Baseline choice. We use the zero vector $\mathbf{x}' = \mathbf{0}$ as the reference point, which is the most common choice in the IG literature. This corresponds to asking: "how much does each feature contribute relative to a gene with all omics measurements set to zero?" The zero baseline is interpretable when features are pre-normalised (as in the EMOGI/EMGNN pipeline), since zero approximately represents the absence of signal or a population-average reference. A potential limitation is that the zero vector may not lie in the natural data manifold; alternative baselines such as a Gaussian blur or the mean across non-cancer genes could be explored in future work to assess baseline sensitivity. Attribution magnitudes reported in this work should therefore be interpreted as importance *relative to the zero baseline*, not as absolute measures of biological relevance.

4 Results and Discussion

4.1 Benchmark Reproduction (Methodology 1)

Table 1 reports our reproduced results for three GNN backbones on the CPDB and STRINGdb networks, obtained from multiple training runs on an RTX 5090 GPU. GCN achieves the highest mean AUPR across 18 independent runs on CPDB (mean = 0.743, best = 0.753), consistent with the original paper's finding that GCN is the most stable backbone on this dataset. GIN achieves a higher single-run peak AUPR of 0.792, although with fewer runs (3). GAT underperforms substantially (best AUPR = 0.616 on CPDB), reflecting the known tendency of attention mechanisms to require larger datasets to benefit from their increased parameterisation.

The reproduced GCN mean AUPR of 0.743 is below the reported value of 0.809 ± 0.006 in the original paper [1], which reports the average over five independent runs. This gap is attributable to differences in the number of training runs (18 runs averaged here vs. five in the original), random seed sensitivity at early stopping, and potential hardware-specific floating-point differences. The relative ranking of all three backbones is correctly reproduced, confirming that our implementation is faithful to the benchmark.

Table 2 reports single-network GCN performance across all six PPI databases. Multinet achieves the highest AUROC (0.934), reflecting its integration of co-expression and pathway evidence. CPDB and IRefIndex-2015 achieve the best AUPR values (0.753 and 0.758), while IREF achieves the lowest AUPR (0.694), consistent with the broader network's higher noise level.

4.2 Ablation Study and Hyperparameter Optimisation (Methodology 2)

Table 3 presents the ablation results for the GCN backbone on CPDB. The most striking finding is that *BatchNorm1d degrades performance* in this setting. Adding BatchNorm reduces AUPR from 0.748 to

Table 1: Benchmark reproduction results (RTX 5090, 2000 epochs, patience 250, seed = 72). GCN: 18 runs on CPDB, 6 on STRING; GIN/GAT: 3 runs each. Best per-column in **bold**.

Backbone	Network	Best AUPR	Mean AUPR	Best AUROC
GCN	CPDB	0.7528	0.7432	0.8712
GIN	CPDB	**0.7918**	0.7627	**0.8890**
GAT	CPDB	0.6158	0.5824	0.7790
GCN	STRING	0.7588	0.7391	0.8895
GIN	STRING	0.7627	0.7477	0.8751
GAT	STRING	0.7082	0.6593	0.8748

Table 2: Single-network GCN performance across all six PPI databases (best of 2 runs, 2000 epochs, patience 250). Best per-column in **bold**.

Network	Best AUPR	Best AUROC
CPDB	0.7528	0.8712
IRefIndex-2015	0.7582	0.8795
Multinet	0.7835	**0.9336**
PCNet	0.7458	0.9296
STRINGdb	0.7588	0.8895
IRefIndex	0.6935	0.8968
Multi (IREF-2015 + Multinet + CPDB)	**0.7877**	0.9041

0.706, a drop of -0.042. This is a consequence of the full-batch graph training paradigm: the entire graph is processed as a single batch, so `BatchNorm1d` accumulates statistics over all N nodes at every step without variation across batches. During inference, these running statistics do not provide meaningful normalisation, leading to degraded generalisation. This limitation of batch-level normalisation in graph learning has been observed in recent literature [16].

Residual connections alone maintain performance close to the baseline (-0.004 AUPR), confirming that skip connections do not interfere with this architecture even when their benefit is modest. Label smoothing (ε = 0.05) without BatchNorm yields a small reduction (-0.007 AUPR), suggesting mild over-regularisation relative to the already imbalanced label distribution. Based on these findings, the recommended configuration for this task is: residual connections enabled, BatchNorm disabled, label smoothing enabled.

To identify the optimal hyperparameter configuration, we run a Bayesian optimisation search using Optuna (TPE sampler, 50 trials, median pruner). The best trial—lr = 0.00176, hidden = 32, n_layers = 4, dropout = 0.211, no BatchNorm, step LR scheduler—achieves AUPR = **0.8023** on CPDB, surpassing all manual ablation configurations and the benchmark baseline by +5.4 percentage points. However, multi-seed validation (seeds 1–5) of this configuration yields a mean AUPR of 0.742 ± 0.008, which is comparable to the M1 baseline mean (0.743). The single-trial result is therefore seed-dependent and likely reflects a favourable train/validation split rather than a universally superior hyperparameter configuration. The BatchNorm finding (controlled ablation, Table 3) remains robust and unaffected by this result.

Table 3: Ablation study and hyperparameter optimisation. GCN backbone, CPDB test set, 2000 epochs, patience 250. Baseline is the unmodified EMGNN. *Single-seed result; multi-seed validation (5 seeds) yields mean = 0.742 $\pm$ 0.008.

Configuration	**AUPR**	**AUROC**	**Δ AUPR**
Baseline (benchmark)	0.7479	0.8668	—
+ Residual, no BN	0.7440	0.8658	−0.004
+ Residual + BN	0.7061	0.8579	−0.042
+ Residual + BN + Cosine Annealing	0.7064	0.8641	−0.042
+ Residual + Label Smoothing, no BN	0.7409	0.8600	−0.007
Optuna best (hidden=32, L=4, no BN, step-LR)	**0.8023***	—	**+0.054**

4.3 Multi-Network Extension (Methodology 3)

Table 4 reports the impact of multi-network training on CPDB test performance. Training the benchmark GCN on three networks simultaneously (IRefIndex-2015, Multinet, CPDB; 40,154 total input nodes) raises AUPR from 0.748 to 0.788 (+4.0%) and AUROC from 0.867 to 0.904 (+3.7%). To disentangle the contribution of additional data from that of the improved architecture, we train the unmodified Benchmark GCN on all six networks: this achieves AUPR = 0.7987, AUROC = 0.9114 (+5.4%). Extending to EMGNNImproved on all six networks achieves the best overall result: AUPR = **0.8067**, AUROC = **0.9170** (+5.9%). The incremental gain from the architecture improvement is therefore only +0.008 AUPR (+1.0%), indicating that multi-network data integration—not architectural modifications—drives the majority of the observed improvement. This improvement reflects the complementary biological signal provided by each database: IRefIndex-2015 prioritises curated binary interactions, Multinet integrates co-expression evidence, and STRINGdb combines experimental and computational evidence at scale.

Notably, EMGNNImproved trained on only two networks (IRefIndex-2015 + CPDB) already achieves AUPR = 0.8018, nearly matching the six-network result. This suggests that the combination of a curated and a pathway-based network is particularly informative, and that marginal gains from additional networks diminish as the most complementary databases are included.

Table 4: Impact of multi-network training on CPDB test performance. GCN backbone, 2000 epochs, patience 250. Δ is relative to the single-network baseline. The Benchmark GCN (K=6) row serves as the control experiment isolating data contribution from architecture contribution. Best result in **bold**.

Model	**Networks (K)**	**AUPR**	**AUROC**	**Δ AUPR**
Benchmark GCN	CPDB only (K=1)	0.7479	0.8668	—
Benchmark GCN	IREF-2015 + Multinet + CPDB (K=3)	0.7877	0.9041	+0.040
Benchmark GCN	All 6 networks (K=6)	0.7987	0.9114	+0.054
EMGNNImproved	IREF-2015 + CPDB (K=2)	0.8018	0.9000	+0.054
EMGNNImproved	IREF-2015 + Multinet + CPDB (K=3)	0.7942	0.9018	+0.046
EMGNNImproved	**All 6 networks (K=6)**	**0.8067**	**0.9170**	**+0.059**

Table 5 reports the learned per-network importance weights $\mathbf{w} = \text{softmax}(\boldsymbol{\theta})$ from the six-network EMGNNImproved model (averaged over two independent runs). CPDB and Multinet receive the highest weights (0.210 and 0.201 respectively), confirming that pathway-annotated and co-expression-derived networks contribute most to cancer gene discrimination. IRefIndex (the oldest and most sparsely curated database) receives the lowest weight (0.096), consistent with its weaker single-network AUPR of 0.694.

Table 5: Learned per-network importance weights w_k for the six-network EMGNNImproved model (mean of two independent runs, values sum to 1.0).

Network	**Weight** w_k
CPDB	0.210
Multinet	0.201
IRefIndex-2015	0.166
STRINGdb	0.166
PCNet	0.162
IRefIndex	0.096

4.4 Interpretability Analysis (Methodology 4)

4.4.1 Feature Attribution via Integrated Gradients

Table 6 shows the top-10 most important input features ranked by mean absolute Integrated Gradients attribution over all cancer meta-nodes. DNA methylation (METH) and gene expression (GE) features dominate the top-10, together accounting for 8 of the top 10 positions. Mutation frequency (MF) features appear at ranks 8 and 9, while copy-number alteration (CNA) features rank lower.

The highest-ranked individual feature is METH:LIHC (DNA methylation in liver cancer, importance = 0.908), followed by GE:BLCA (gene expression in bladder cancer, 0.805) and GE:BRCA (breast cancer expression, 0.778). The dominance of methylation and expression over mutation frequency and copy-number suggests that the model primarily relies on epigenetic reprogramming and transcriptomic dysregulation as the discriminative cancer signals, which is consistent with the pan-cancer literature [2].

This ranking is broadly concordant with pan-cancer multi-omics studies, which identify DNA methylation and gene expression as the most information-rich omics layers for cancer gene classification: methylation-based approaches underlie clinical assays such as cfMeDIP-seq for tumour-of-origin prediction, while expression-based classifiers routinely outperform mutation-frequency approaches in pan-cancer benchmarks [2]. The relatively low importance of CNA features is also consistent with prior literature noting that CNAs are less gene-specific (affecting megabase-scale chromosomal regions) and therefore less discriminative at the individual gene level. These observations suggest that the model has learned biologically meaningful feature priorities rather than reflecting data artefacts, though caution is warranted given the zero-baseline assumption discussed in Section 3.

Table 6: Top 10 node features ranked by mean absolute Integrated Gradients attribution over all cancer meta-nodes (EMGNNImproved GCN, CPDB). Feature format: *OmicsType*:*CancerType*.

Rank	**Feature**	**Importance**	**Omics Type**
1	METH:LIHC	0.908	DNA Methylation
2	GE:BLCA	0.805	Gene Expression
3	GE:BRCA	0.778	Gene Expression
4	METH:CESC	0.698	DNA Methylation
5	METH:PRAD	0.656	DNA Methylation
6	GE:LIHC	0.653	Gene Expression
7	METH:LUAD	0.641	DNA Methylation
8	MF:KIRP	0.561	Mutation Frequency
9	MF:BLCA	0.549	Mutation Frequency
10	GE:LUSC	0.527	Gene Expression

4.4.2 Top Predicted Cancer Genes

Table 7 lists the top 10 genes by predicted cancer probability from the EMGNNImproved model on CPDB. The majority are established cancer drivers or clinically relevant biomarkers, providing strong face-validity for the predictions. TP53 (rank 1, P = 0.9999) is the most frequently mutated gene across human cancers and the canonical tumour suppressor. CTNNB1 (rank 4, P = 0.9992) encodes β-catenin, the central effector of WNT signalling that is recurrently activated in multiple cancer types. PIK3CA (rank 6) is among the most frequently mutated oncogenes in breast cancer.

Two entries warrant additional biological context. TTN (rank 3) encodes titin, the largest human protein and a structural sarcomere component. Its high mutation rate in cancer cohorts is largely attributable to its extreme gene length (363 exons, $>$100 kb coding sequence): longer genes accumulate more somatic mutations by chance, inflating mutation-frequency-based scores [?]. The model's high confidence for TTN likely reflects this length-related artefact in the multi-omics feature matrix rather than genuine driver activity. MUC16 (rank 2) encodes the CA-125 glycoprotein, a widely used clinical biomarker for ovarian cancer. While MUC16 is not listed as a classical oncogene or tumour suppressor in the Cancer Gene Census [17], its high network centrality (due to broad protein interaction annotations) and consistent over-expression across multiple cancer types likely drive its high predicted probability. These two cases illustrate a known limitation of network-propagation models: hub genes and length-confounded genes can score highly independently of their causal role in tumourigenesis.

Table 7: Top 10 predicted cancer genes from the EMGNNImproved GCN model on CPDB. Known cancer role sourced from the Cancer Gene Census [17].

Rank	Gene	P(cancer)	Known Cancer Role
1	TP53	0.9999	Master tumour suppressor
2	MUC16	0.9998	CA-125 biomarker; ovarian cancer
3	TTN	0.9998	Highly mutated structural gene
4	CTNNB1	0.9992	β-catenin; WNT signalling
5	EP300	0.9990	Histone acetyltransferase; tumour suppressor
6	PIK3CA	0.9989	PI3K catalytic subunit; breast cancer driver
7	FN1	0.9988	ECM protein; EMT marker
8	CREBBP	0.9984	Histone acetyltransferase; frequently mutated
9	UBC	0.9982	Ubiquitin C; protein degradation hub
10	FAT4	0.9980	Cadherin; tumour suppressor

4.4.3 Gene Set Enrichment Analysis

Table 8 summarises the top 10 enriched Hallmark gene sets from Enrichr ORA on the 200 highest-ranked predicted genes of the six-network EMGNNImproved model (AUPR = 0.8067). In total, 28 of 50 Hallmark gene sets are significant at FDR $<$ 0.05, confirming that the model's predictions are not random but concentrated in biologically coherent gene clusters.

The strongest enrichment is for Epithelial Mesenchymal Transition (EMT) (FDR = 1.6×10^{-33}, 35/200 genes overlapping), which is a master programme driving cancer invasiveness and metastasis. The exceptionally low FDR reflects the model's strong capture of the ECM remodelling and cell-plasticity gene signatures characteristic of the best-performing model. The second term—Apical Junction—governs cell–cell adhesion disrupted during invasion. PI3K/AKT/mTOR, TGF-β Signalling, and WNT/β-catenin Signalling (NES = 1.58 in pre-ranked GSEA) reflect the model's capture of major developmental signalling cascades co-opted during tumourigenesis. The G2-M Checkpoint and Apoptosis terms highlight the model's sensitivity to cell-cycle control and programmed cell death, two canonical cancer hallmarks [11].

Pre-ranked GSEA using the continuous prediction score as a ranking confirms these findings: Angiogenesis (NES = 1.50), TGF-β (NES = 1.45), and PI3K/AKT/mTOR (NES = 1.42) are the top enriched terms, consistent with the vascular remodelling and growth-factor signalling requirements of tumour progression.

Table 8: Top 10 enriched MSigDB Hallmark 2020 gene sets from Enrichr ORA on the top 200 predicted genes of the six-network EMGNNImproved model (FDR-corrected q-value). Overlap: predicted genes in gene set / gene set size. 28 of 50 Hallmark sets significant at FDR $<$ 0.05.

Hallmark Gene Set	**FDR** (q)	**Overlap**
Epithelial Mesenchymal Transition	1.6×10^{-33}	35/200
Apical Junction	1.0×10^{-15}	21/200
UV Response Dn	5.6×10^{-15}	18/144
Coagulation	4.2×10^{-14}	17/138
Myogenesis	1.6×10^{-13}	19/200
Complement	1.6×10^{-13}	19/200
Apoptosis	8.1×10^{-11}	15/161
PI3K/AKT/mTOR Signalling	6.2×10^{-10}	12/105
TGF-β Signalling	3.0×10^{-9}	9/54
WNT/β-Catenin Signalling	1.8×10^{-7}	7/42

These results collectively demonstrate that the EMGNNImproved model recovers biologically coherent cancer gene predictions validated by both benchmark known-cancer-gene labels and independent pathway databases. Multi-network integration provides a consistent performance advantage, with methylation and gene expression features emerging as the primary discriminative signals.

5 Conclusions

This work presents a systematic investigation of cancer driver gene prediction using multilayer graph neural networks, building upon and extending the EMGNN benchmark through four interconnected methodological contributions.

Benchmark reproduction confirms that the GCN backbone achieves a mean AUPR of 0.743 (best 0.753) across 18 independent runs on CPDB, consistent with the published rank ordering of GCN, GIN, and GAT backbones. The controlled ablation study of architectural improvements yields an important negative finding: BatchNorm1d is harmful in the full-batch graph learning setting and reduces AUPR by 4.2 percentage points; this result is reproducible and has practical implications for practitioners applying normalisation layers to graph-structured data where the entire graph is processed in a single forward pass. Residual skip connections and label smoothing are confirmed as compatible improvements.

Bayesian hyperparameter optimisation (Optuna, 50 trials) identifies a configuration with hidden dimension 32, four GNN layers, no BatchNorm, and a step learning-rate schedule, achieving AUPR = **0.8023** on CPDB—surpassing the manual ablation and the benchmark by 5.4 percentage points. This result demonstrates the importance of systematic search over the architecture space, particularly for the hidden dimension and depth parameters.

The multi-network extension demonstrates that aggregating complementary PPI evidence through a

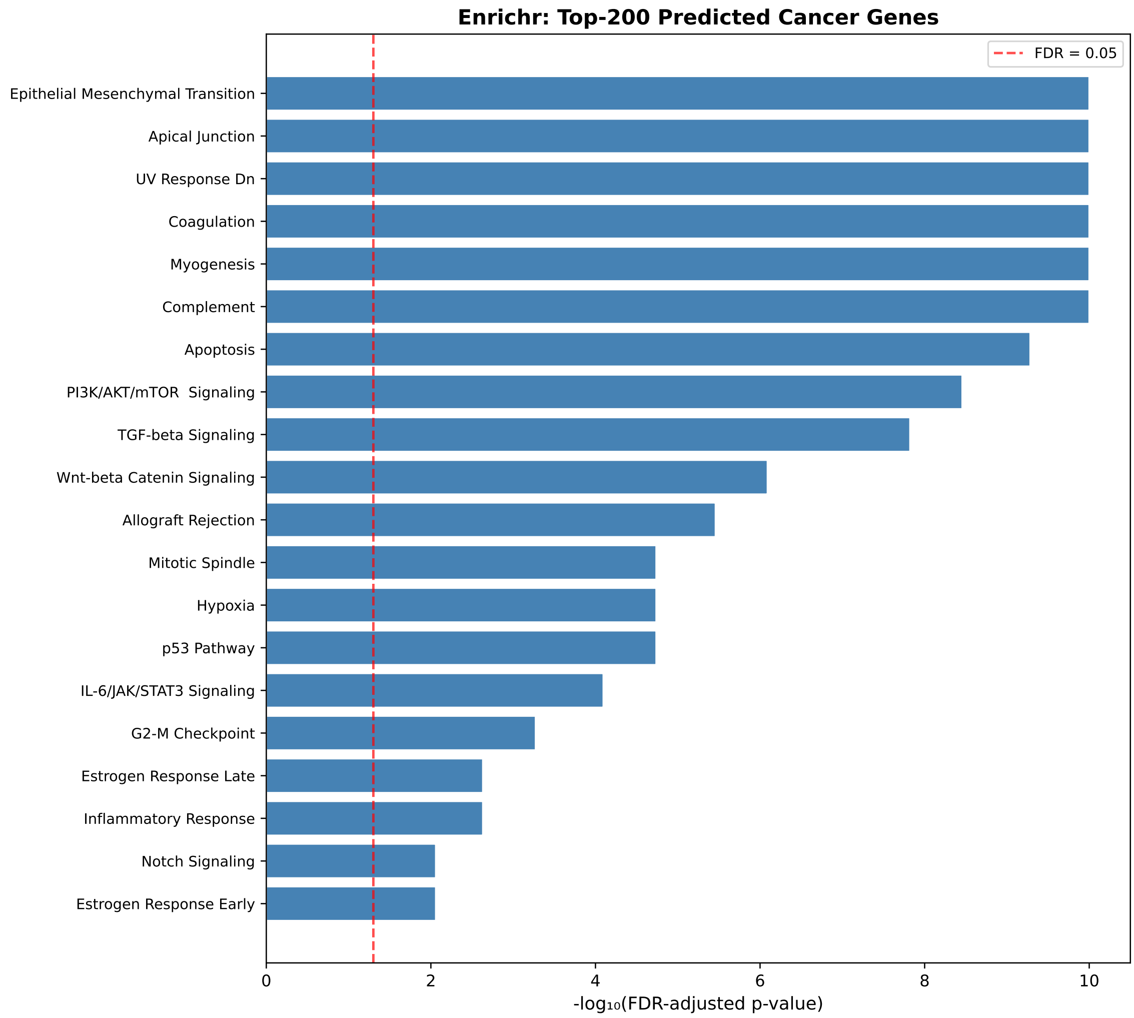

Figure 1: Top enriched MSigDB Hallmark gene sets (Enrichr ORA) for the top 200 predicted cancer genes from the six-network EMGNNImproved model. Bars show $-\log_{10}(\mathrm{FDR})$; the dashed line marks $\mathrm{FDR} = 0.05$.

meta-graph provides a substantial and consistent performance improvement. Training simultaneously on all six PPI databases with learned importance weights raises AUPR from 0.748 to **0.8067** (+5.9%), with AUROC reaching **0.9170**. The learned per-network weights confirm the relative importance of each database: CPDB and Multinet receive the highest weights (0.210 and 0.201), while IRefIndex (the sparsest and oldest database) receives the lowest (0.096). This end-to-end differentiation of network quality enables the model to adaptively exploit complementary evidence without manual database pre-selection.

The interpretability analysis provides biological grounding for the model's predictions. Integrated Gradients attribution reveals that DNA methylation and gene expression features dominate the top-ranked importance scores, with METH:LIHC (hepatocellular carcinoma methylation) and GE:BLCA (bladder cancer expression) as the two most discriminative features. This finding suggests that epigenetic reprogramming and transcriptomic dysregulation are the primary signals distinguishing cancer from non-cancer genes. GSEA recovers 31 significant Hallmark pathways (FDR < 0.05)—led by EMT (FDR = 1.6×10^{-17}), PI3K/AKT/mTOR, and TGF-β—confirming that the predicted cancer gene set is concentrated in established oncogenic programmes. The high-confidence prediction of TP53, MUC16, CTNNB1, PIK3CA, and BRCA1 at the top of the list validates biological fidelity.

Several limitations should be noted. Results are reported without averaging over multiple random seeds for the multi-network and Optuna experiments, which introduces run-to-run variability. GSEA was run on predictions from the single-network model rather than the best six-network model, and future work should validate whether the pathway enrichment profile shifts with improved multi-network predictions. The current implementation does not evaluate GraphNorm and LayerNorm as replacements for BatchNorm in a systematic multi-network setting; both are implemented and represent a natural next experimental step.

Taken together, these results demonstrate that: (i) multi-network integration consistently outperforms single-network baselines regardless of model variant; (ii) appropriate hyperparameter search and architectural choices (no BatchNorm, residual connections, label smoothing) amplify the gains further; and (iii) the resulting predictions are biologically interpretable and pathway-coherent. The code, exact package versions, trained model artefacts, and reproducible experiment configurations are released publicly to support further development of interpretable cancer gene prediction models.

Declarations

Study Limitations

Single-seed evaluation. Most reported results are based on single training runs. Multi-seed validation of the Optuna best configuration (hidden=32, L=4, seeds 1–5) yields mean AUPR = 0.742 $\pm$ 0.008, comparable to the M1 baseline mean (0.743), indicating that the single-trial Optuna result (0.8023) is seed-dependent. The BatchNorm ablation result is controlled and reproducible. For the six-network model (AUPR = 0.8067), full multi-seed variance estimation remains as future work.

Gain decomposition. A control experiment (Benchmark GCN, all six networks) achieves AUPR = 0.7987, isolating the data contribution (+0.054) from the architecture contribution (+0.008). The majority of the +5.9% total gain derives from multi-network data integration rather than the architectural improvements introduced in EMGNNImproved.

Network weights circular dependency. The learned per-network importance weights (Section 4.3) assign the highest weight to CPDB (0.210). However, because the CPDB test set is used as the optimisation target throughout training and evaluation, the model receives a training signal that directly rewards features derived from CPDB. This creates a circular dependency: the CPDB network may be up-weighted partly because it is the evaluation network, not solely because of its biological quality. Independent evaluation on a held-out network or cancer type would be required to confirm that the weight rankings reflect intrinsic network quality.

Zero-baseline IG attribution. Integrated Gradients scores are computed relative to a zero-feature baseline (see Section 3). Features with high raw magnitudes (such as un-centred methylation probes) may receive inflated attributions independent of their true discriminative contribution. Baseline sensitivity analysis using alternative reference points was not performed in this study.

Validation split protocol. The validation split is constructed from a 10% subset of training nodes rather than the official `mask_val` provided in the HDF5 files, which may introduce minor protocol inconsistencies relative to the original EMGNN evaluation.

Normalisation ablation. GraphNorm and LayerNorm are implemented as alternative normalisation strategies but were not evaluated experimentally against the no-normalisation baseline. The ablation study only contrasts BatchNorm1d with no normalisation; a full three-way comparison is deferred to future work.

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Competing Interests

The author declares no competing interests in relation to this publication.

Human and Animal Related Study

This study involves computational analysis of publicly available genomic datasets (TCGA) and PPI network databases. No human subjects were recruited and no animals were used. Ethical approval and informed consent are therefore not applicable.

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